SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

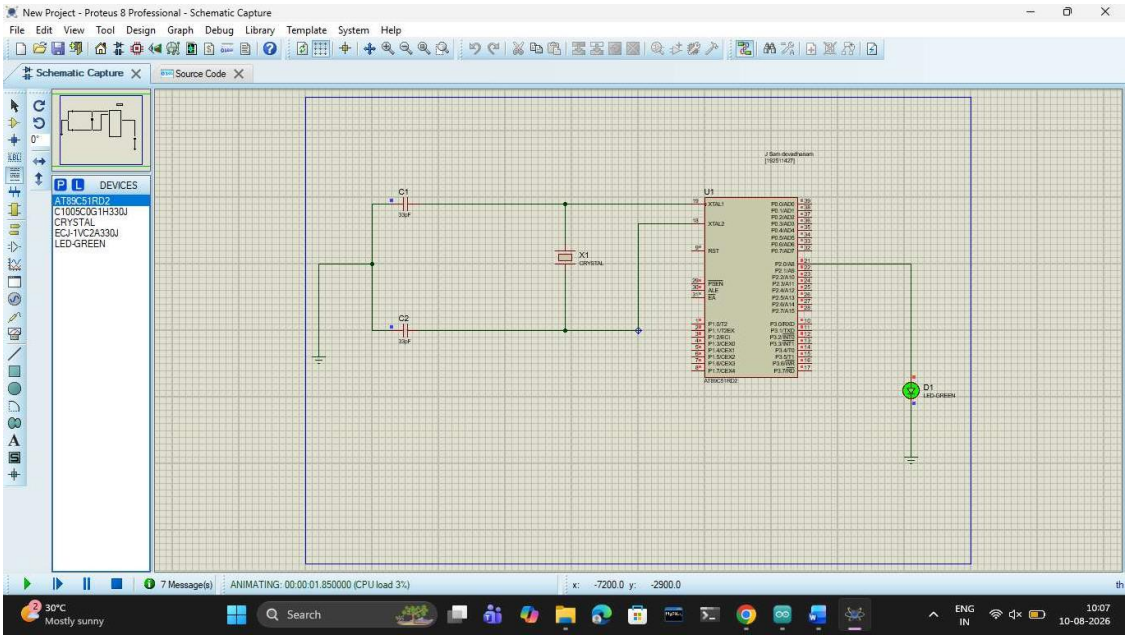
**ECA1423 – EMBEDDED SYSTEMS**

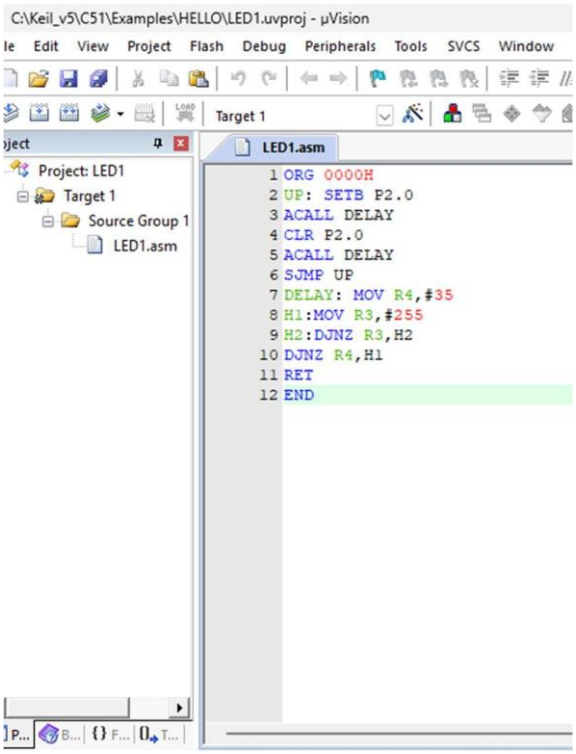
**NAME : SURENDHAR G**
**REG. NO: 192513079**

# LAB EXPERIMENT OUTPUT

# EXP NO: 1

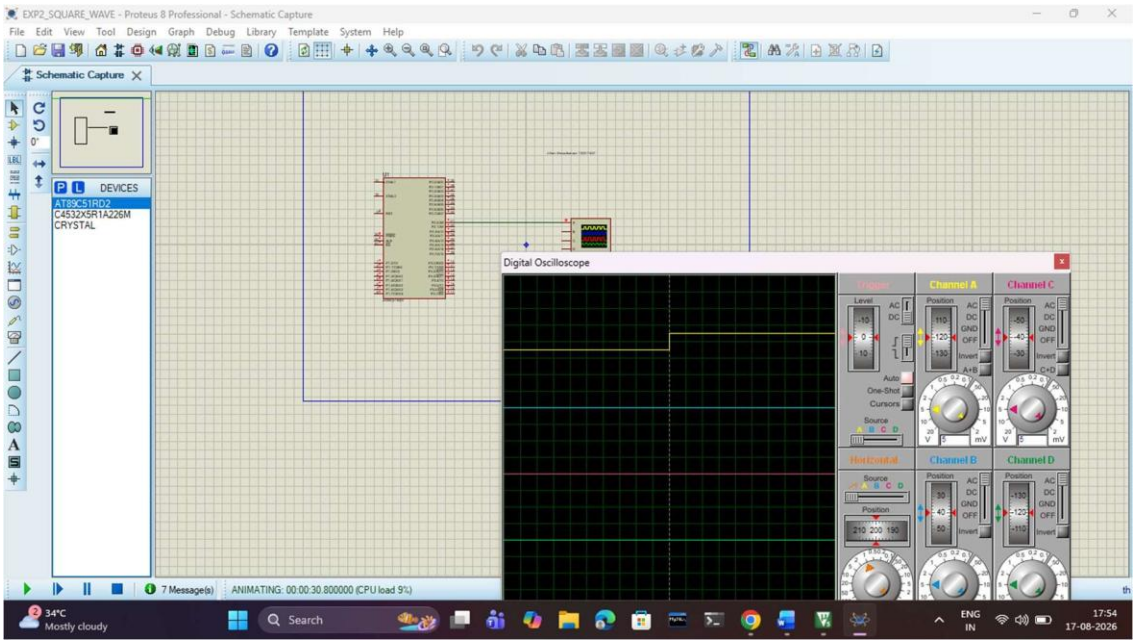
# FLASHING OF LED USING AT89C51 MICROCONTROLLER USING PROTEUS



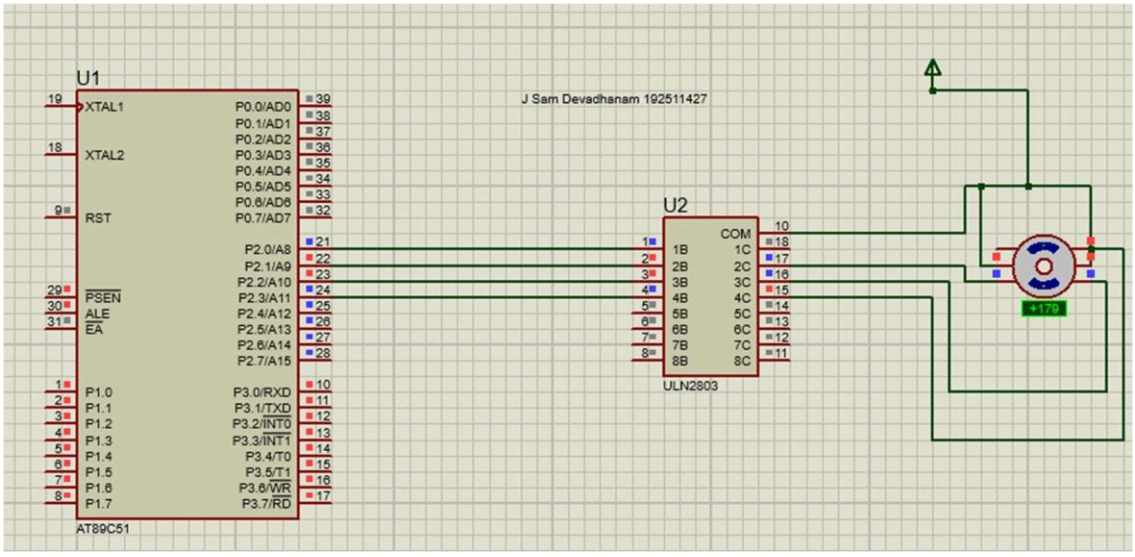
# EXP NO: 2

# GENERATION OF SQUARE WAVE USING PROTEUS



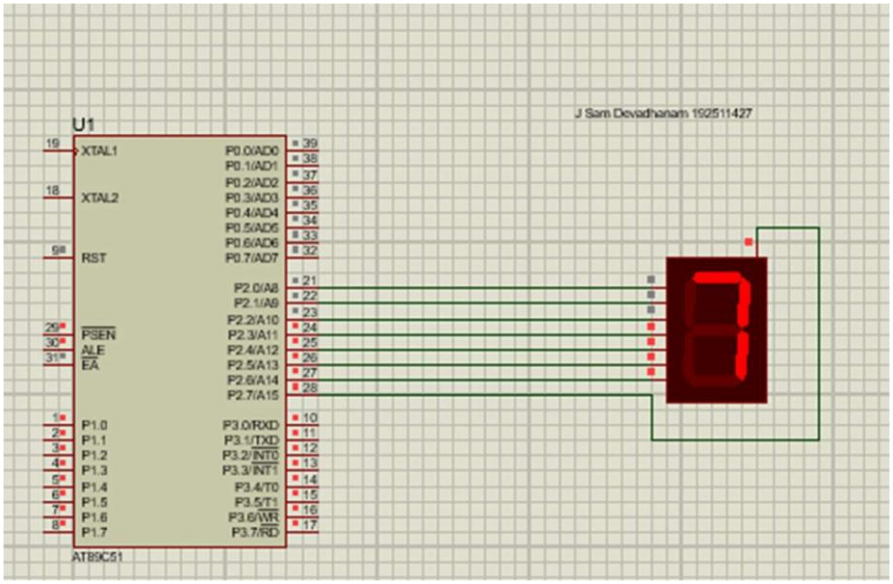
## EXP NO: 3

# STEPPER MOTOR USING AT89C51 USING PROTEUS



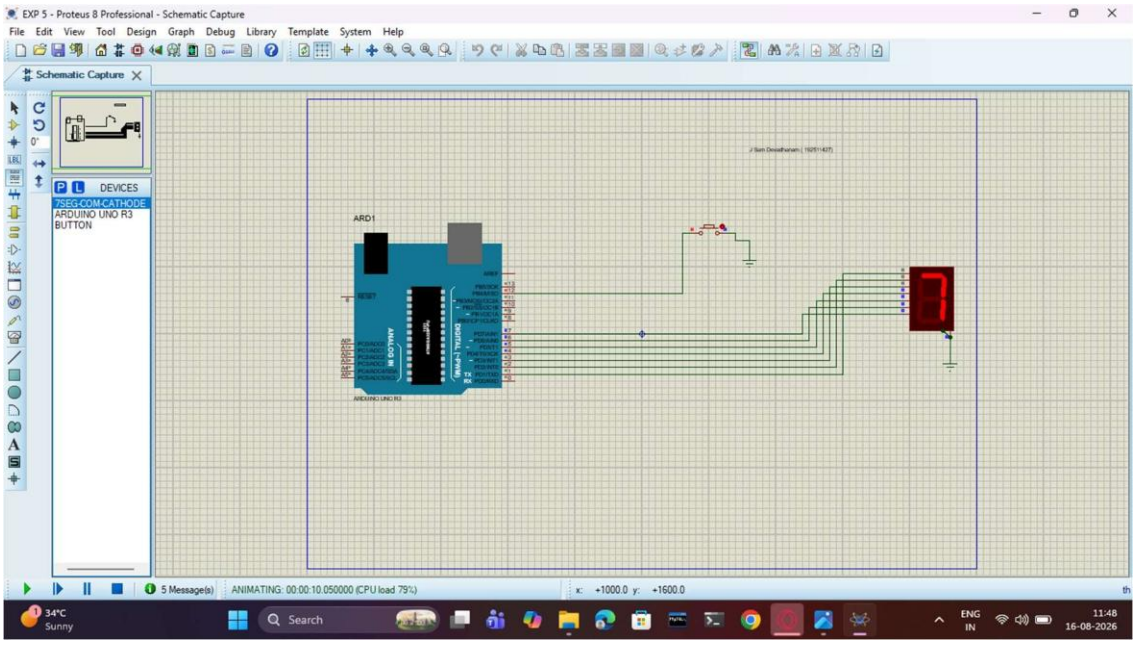
## EXP NO: 4

# 7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS



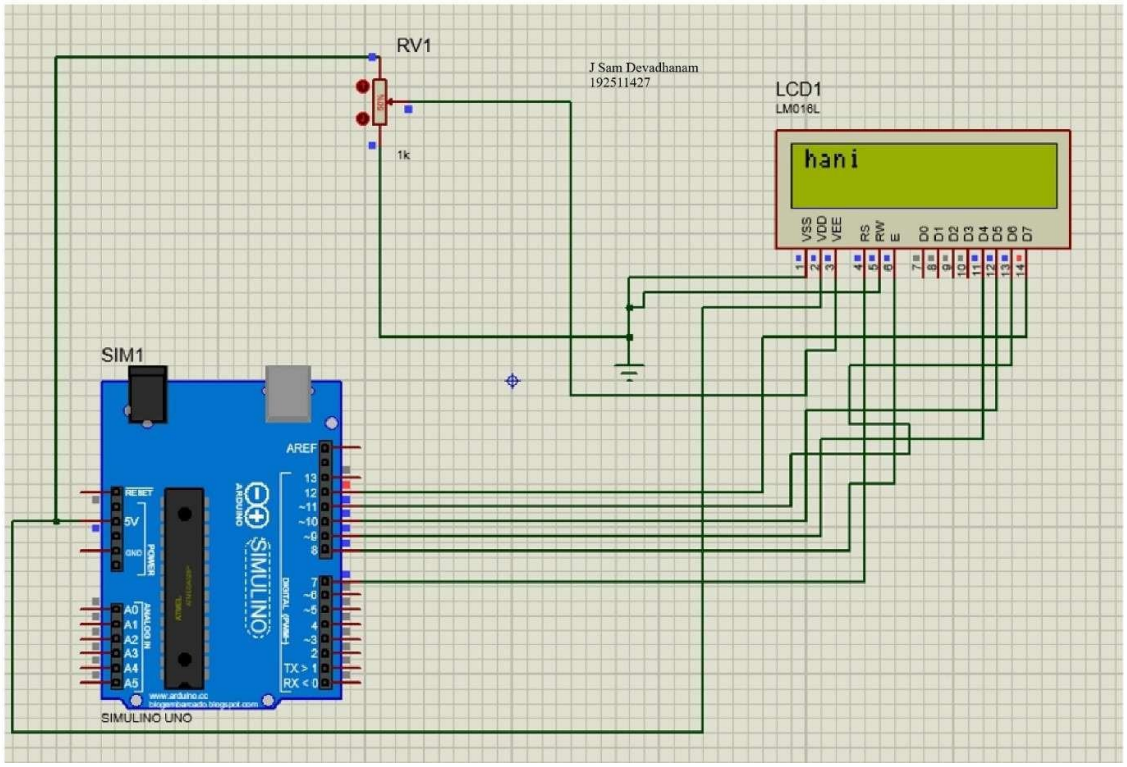
# EXP NO: 5

# ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS



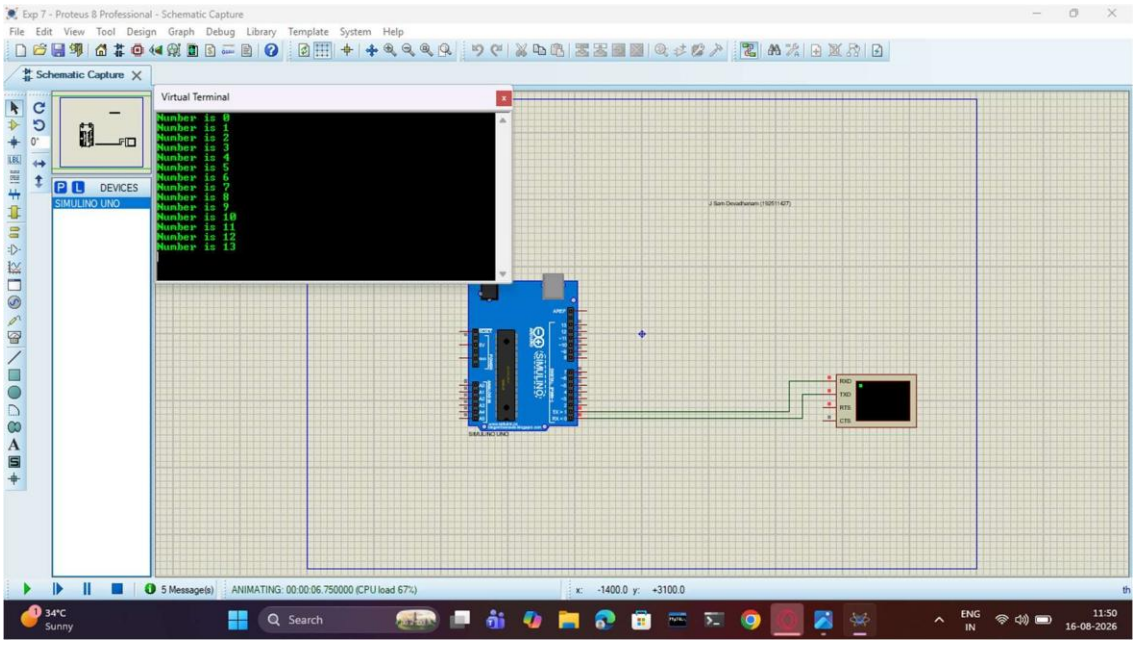
## EXP NO: 6

# ARDUINO TO 16x2 LCD DISPLAY USING PROTEUS



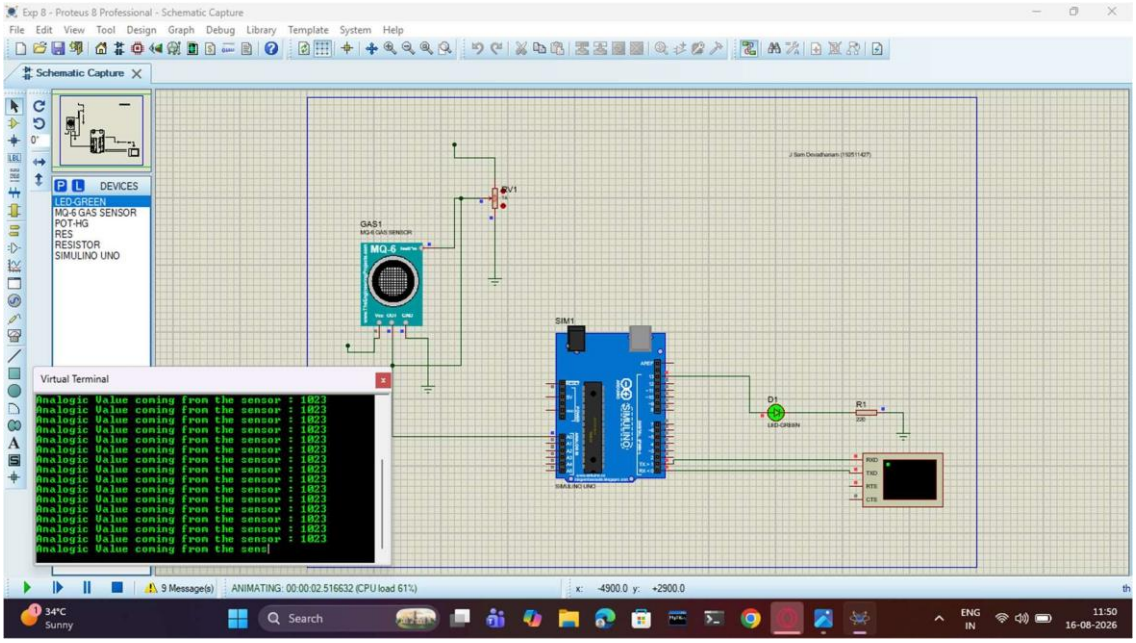
# EXP NO: 7

# SERIAL COMMUNICATION USING ARDUINO WITH PROTEUS



# EXP NO: 8

# GAS SENSOR MQ-2 IN PROTEUS USING ARDUINO

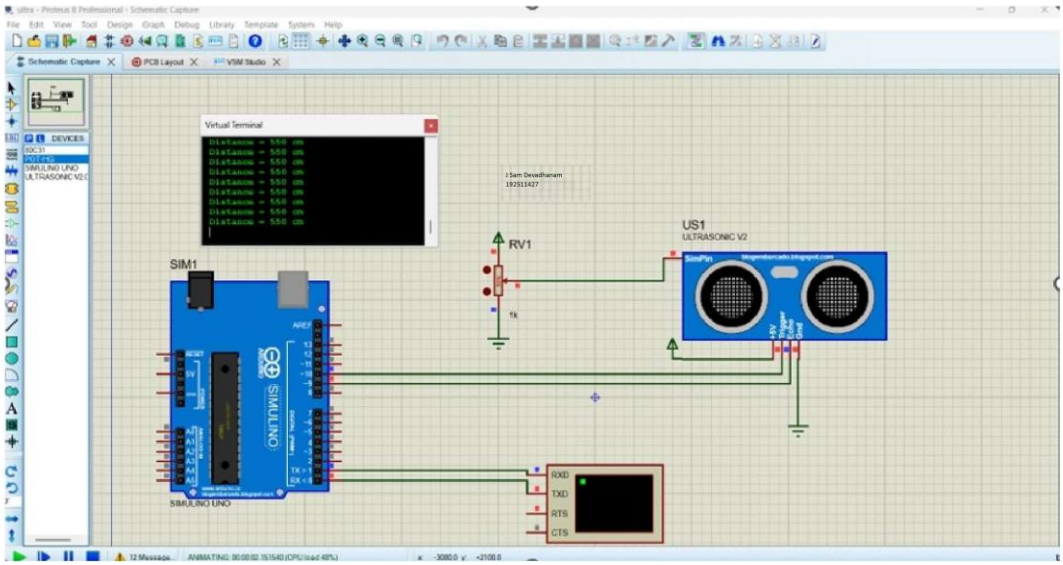


# EXP NO: 9

# ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

*AIM:*

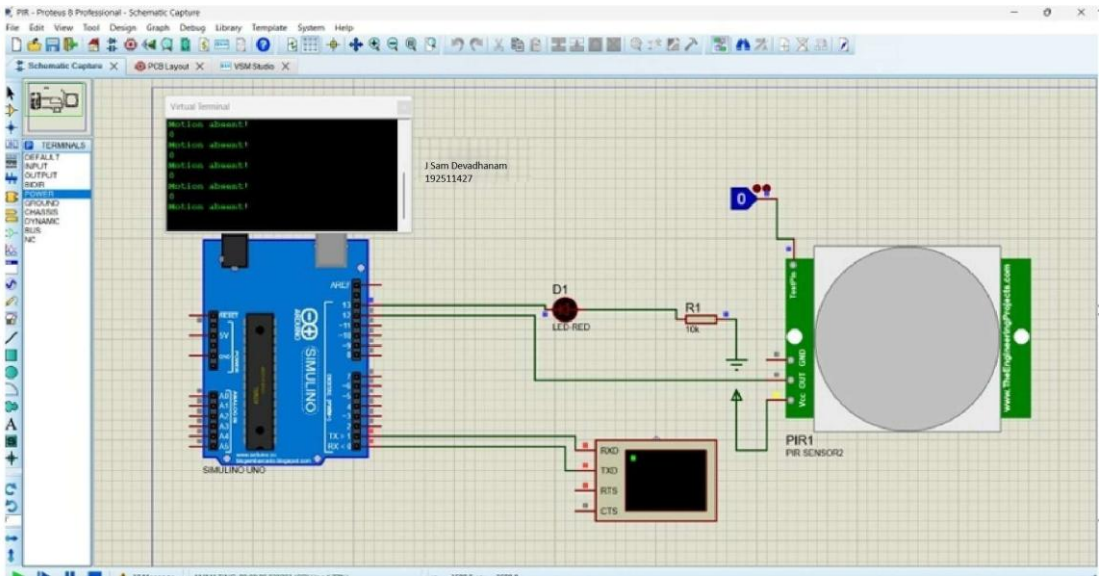
*To write an Embedded C program for interfacing Ultrasonic Sensor using Arduino Uno and Proteus*



# Exp No: 10
# PIR SENSOR IN PROTEUS USING ARDUINO

AIM:

To write an Embedded C program for interfacing PIR Sensor using Arduino Uno and Proteus.

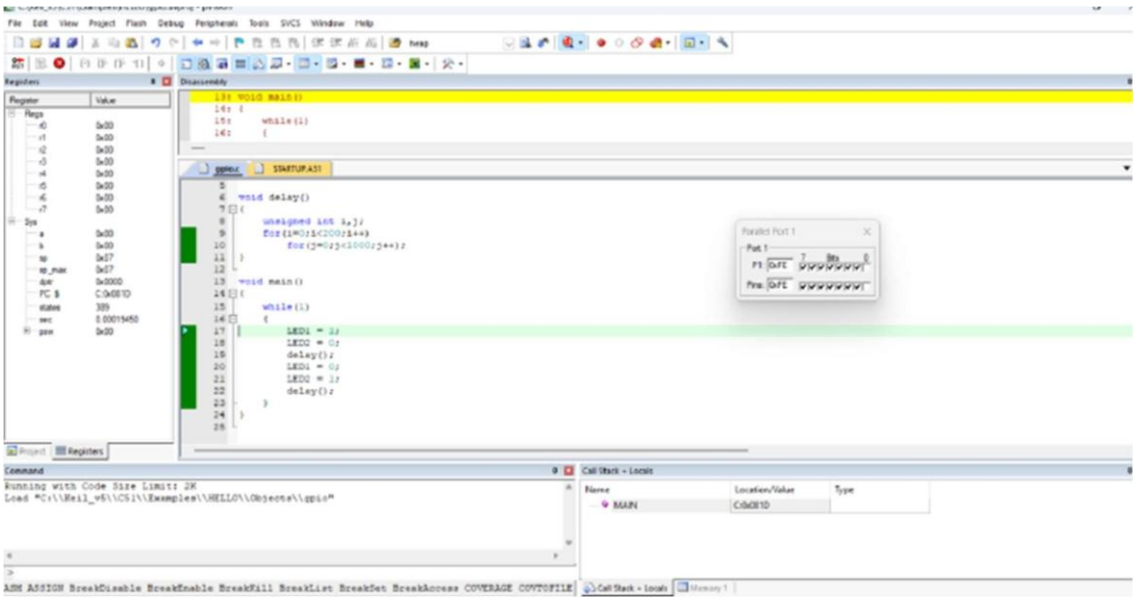


# Exp No: 11

# STUDY OF GPIO & BIT ADDRESSING IN AT89C51

AIM:

To study GPIO control and bit-addressable SFR using Embedded C.



**EXP NO: 12**

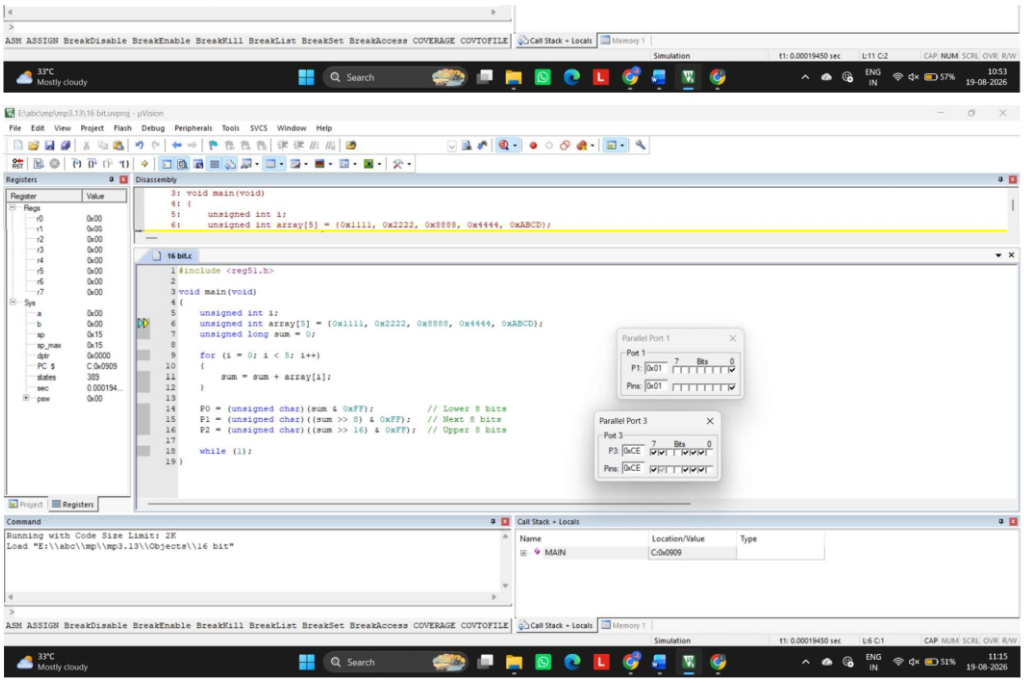
# EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

AIM:

TO WRITE AN EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT NUMBERS USING KEIL SOFTWARE

SOFTWARE REQUIRED:

• KEIL



**EXP NO: 13**

# EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND

# STORE THE 32-BIT RESULT IN INTERNAL RAM

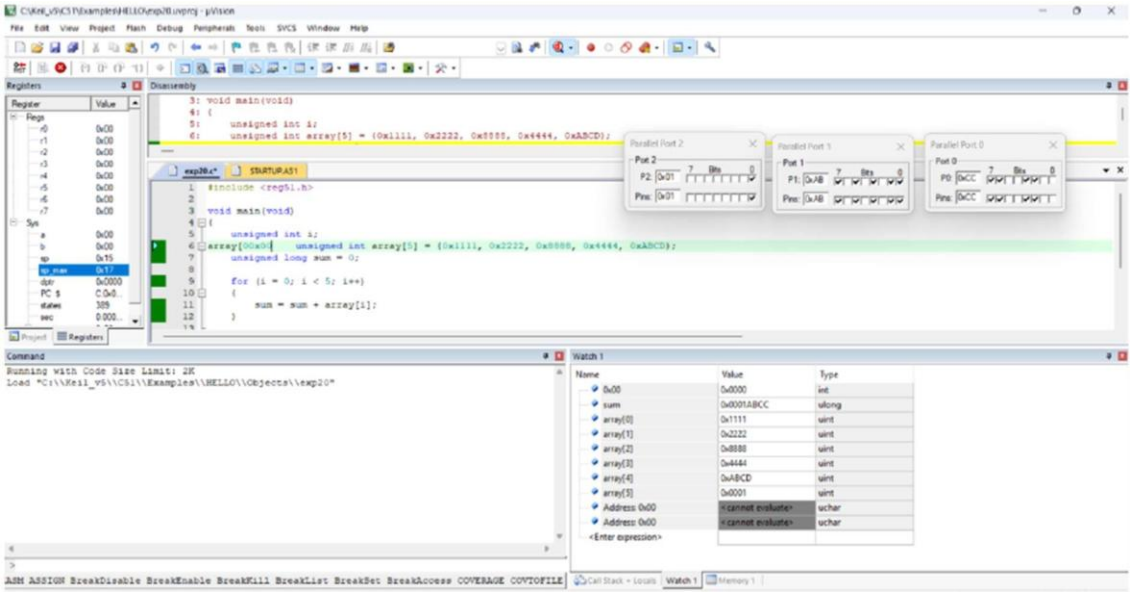
AIM:

TO WRITE AN EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE RESULTS IN

INTERNAL RAM USING KEIL SOFTWARE.

SOFTWARE REQUIRED:

• KEIL



**EXP NO: 14**

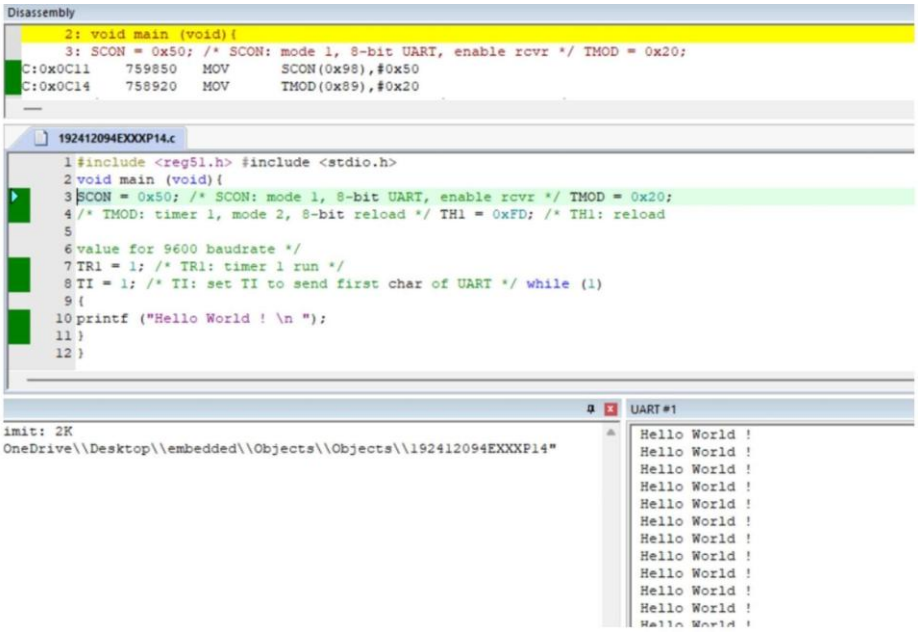
**EMBEDDED C PROGRAM TO DISPLAY "HELLO WORLD" MESSAGE**

AIM:

TO WRITE AN EMBEDDED C PROGRAM TO DISPLAY "HELLO WORLD" MESSAGE USING KEIL SOFTWARE.

SOFTWARE REQUIRED:

• KEIL SOFWARE



**EXP NO: 15**

## EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2

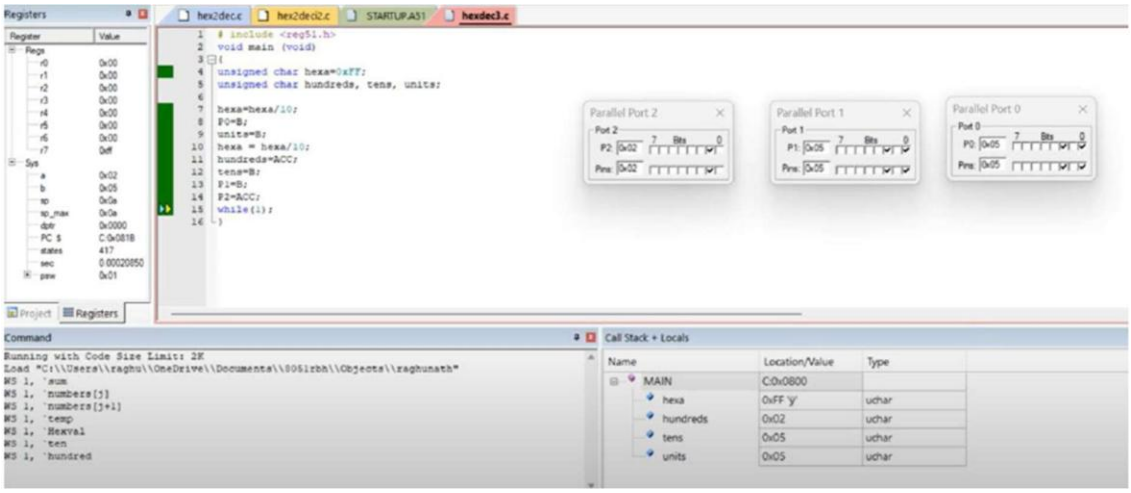
AIM:

TO WRITE AN EMBEDDED C PROGRAM TO CONVERT HEXADECIMAL NUMBERS INTO DECIMAL AND

DISPLAY THE DIGITS ON PORTS USING KEIL SOFTWARE.

SOFTWARE REQUIRED:

• KEIL SOFTWARE



**EXP NO: 16**

# SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

AIM:

TO WRITE AND SIMULATE AN EMBEDDED C PROGRAM USING SERIAL INTERRUPT IN 8051

MICROCONTROLLER SUCH THAT WHILE A MESSAGE IS CONTINUOUSLY TRANSMITTED, ANY KEY PRESSED FROM

THE KEYBOARD GENERATES AN INTERRUPT AND IS DISPLAYED THROUGH UART.

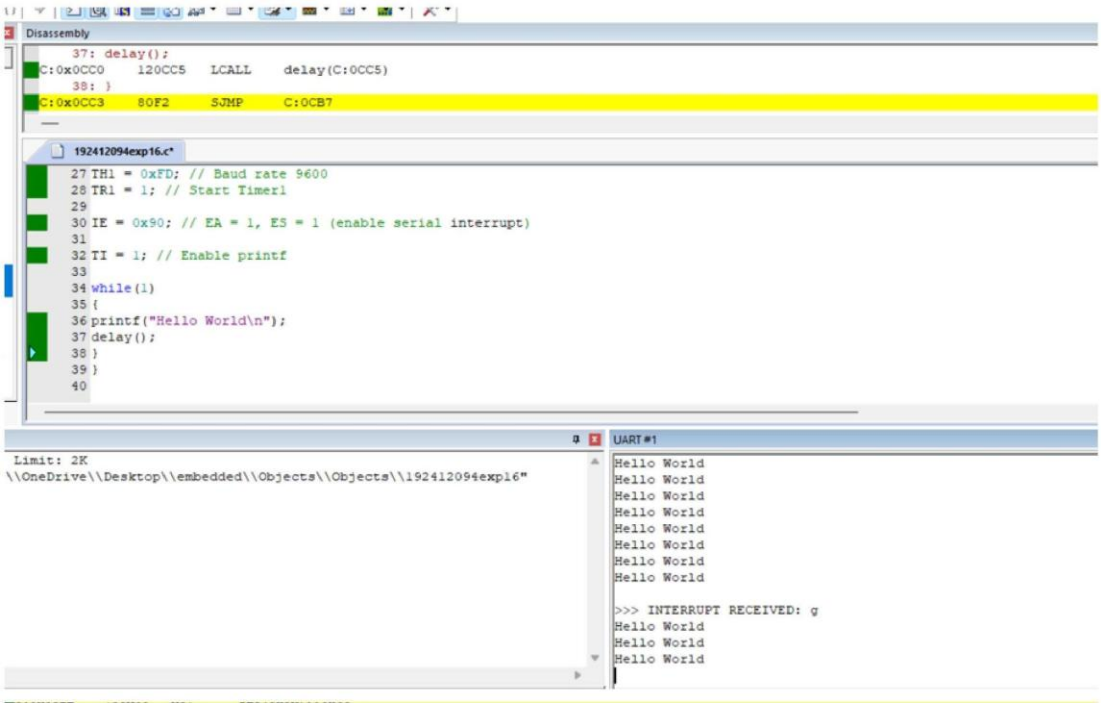
APPARATUS / SOFTWARE REQUIRED:

• PC WITH WINDOWS OS

• KEIL MVISION IDE

• 8051 SIMULATOR (KEIL BUILT-IN)

• SERIAL WINDOW (UART #1)



**EXP NO: 17**

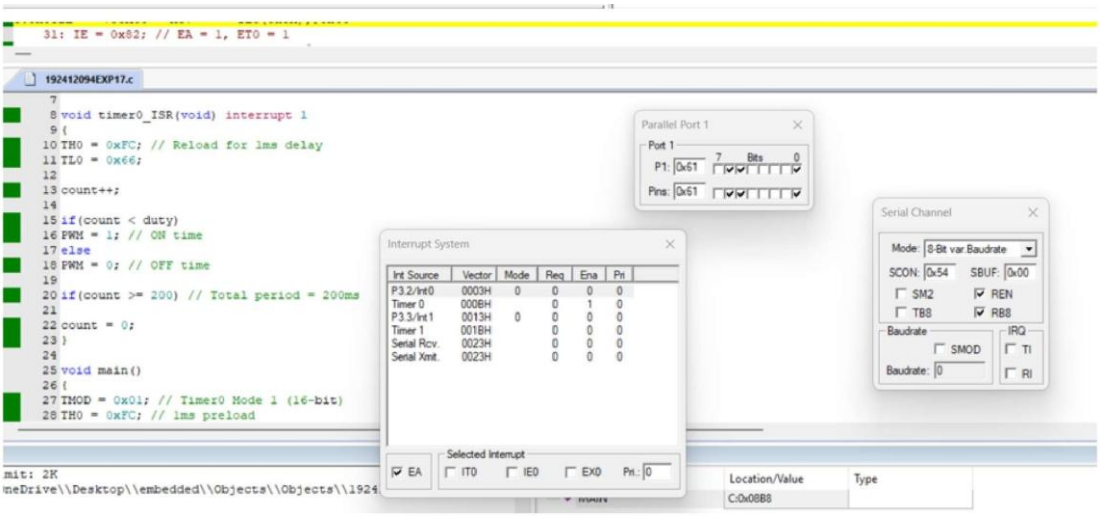
## UART INTERFACING USING INTERRUPT IN KEIL

AIM:

TO RECEIVE SERIAL DATA USING INTERRUPT.

APPARATUS REQUIRED:

- PC WITH WINDOWS OS
- KEIL MVISION IDE
- 8051 SIMULATOR (KEIL BUILT-IN)
- SERIAL WINDOW (UART #1)



**EXP NO: 18**

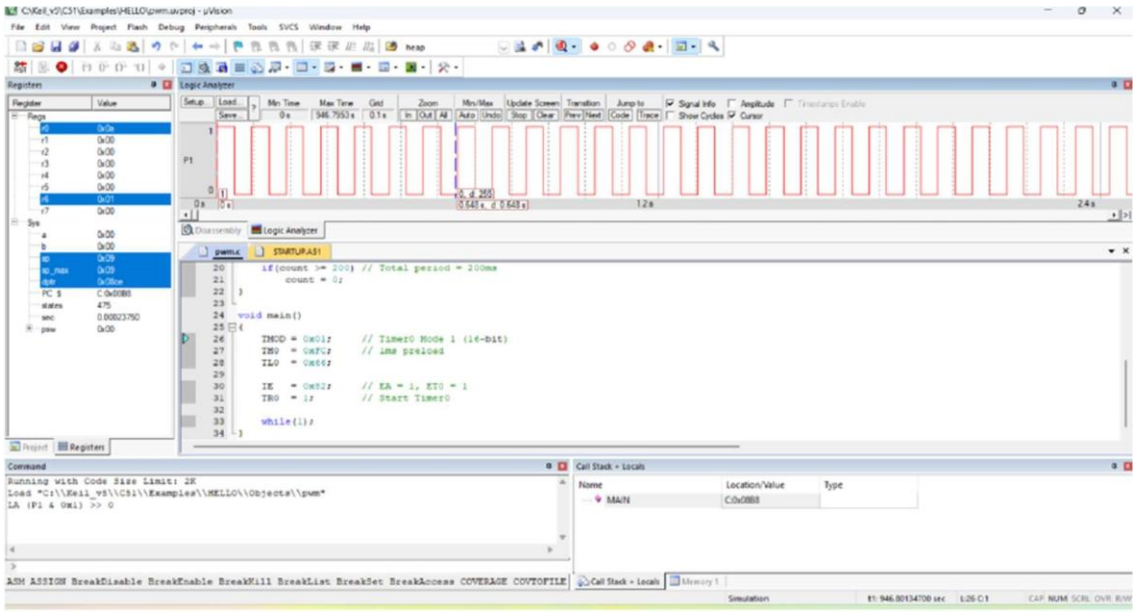
**GENERATION OF PWM USING TIMER INTERRUPT**

AIM:

TO GENERATE PWM USING TIMER INTERRUPT.

APPARATUS / SOFTWARE REQUIRED:

- PC WITH WINDOWS OS
- KEIL MVISION IDE
- 8051 SIMULATOR (KEIL BUILT-IN)
- SERIAL WINDOW (UART #1)



**EXP NO: 19**

## BLINKING LEDS USING SOFTWARE DELAY ROUTINE IN LPC2148 KIT

AIM:

TO WRITE AND EXECUTE A C PROGRAM TO BLINK LEDS USING SOFTWARE DELAY ROUTINE IN LPC

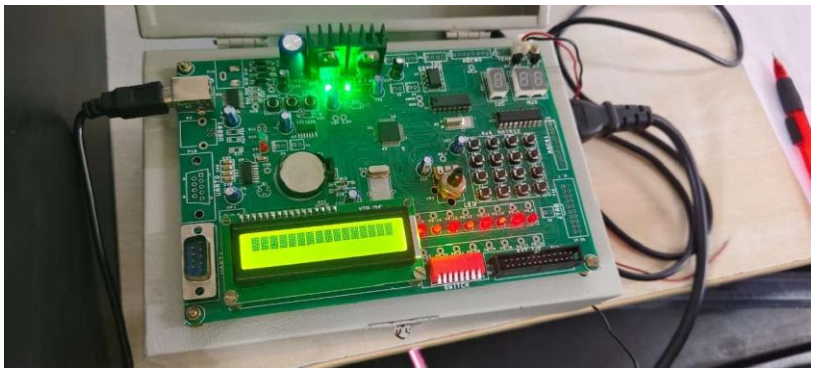
2148 KIT

APPARATUS REQUIRED:

KEIL UVISION5 SOFTWARE

PHILIPS FLAH PROGRAMMER

LPC 2148 KIT



**EXP NO: 20**

## PROGRAM TO READ THE SWITCH AND DISPLAY IN THE LEDS USING LPC2148 KIT

AIM:

TO WRITE AND EXECUTE C PROGRAM TO READ THE SWITCH AND DISPLAY IN THE LEDS USING

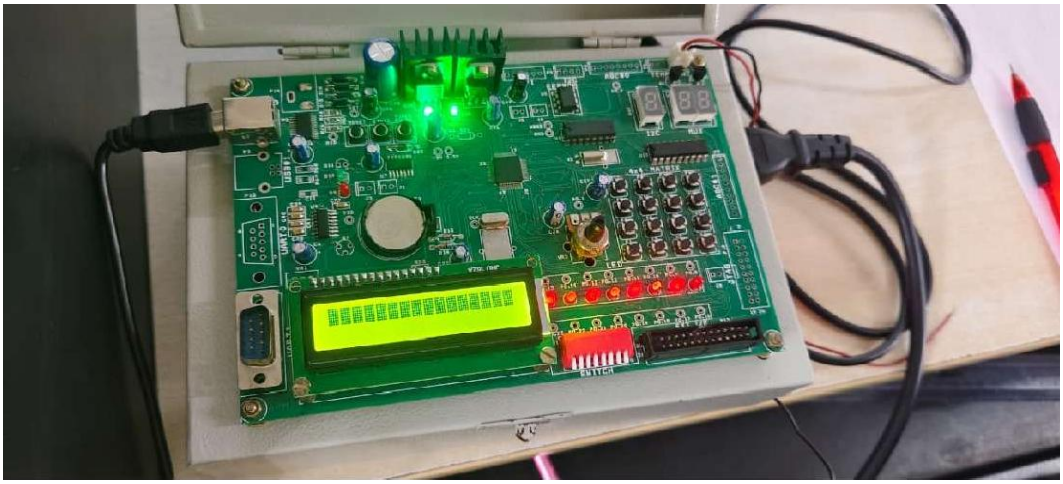
LPC2148 KIT

APPARATUS REQUIRED:

KEIL UVISION5 SOFTWARE

PHILIPS FLAH PROGRAMMER

LPC 2148 KIT



**EXP NO: 21**

**DISPLAY A NUMBER IN SEVEN SEGMENT LED IN LPC2148 KIT**

AIM:

TO WRITE AND EXECUTE C PROGRAM TO DISPLAY A NUMBER IN SEVEN SEGMENT LED IN

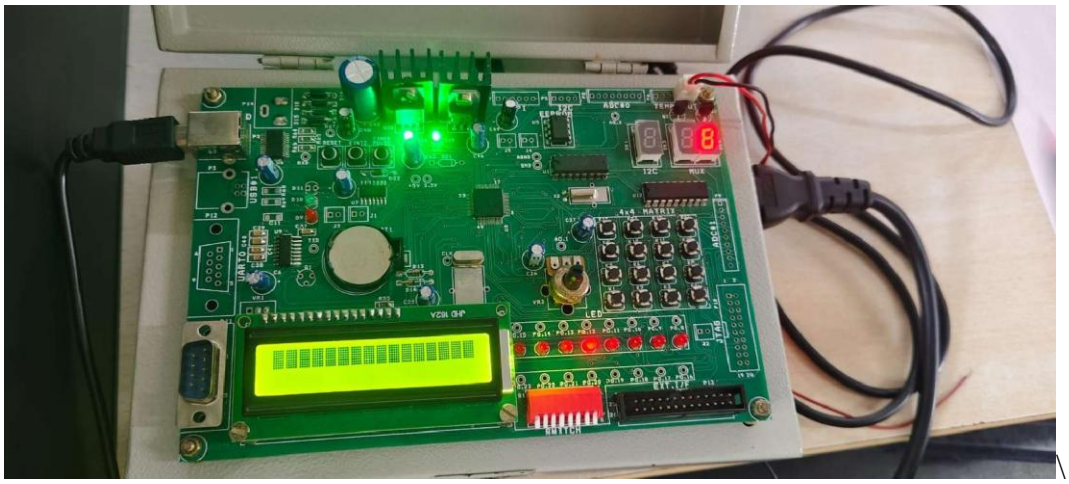
LPC2148 KIT

APPARATUS REQUIRED:

KEIL UVISION5 SOFTWARE

PHILIPS FLAH PROGRAMMER

LPC 2148 KIT



\

**EXP NO: 22**

## SERIAL TRANSMISSION AND RECEPTION USING ON-CHIP UART IN LPC2148 KIT

AIM:

TO WRITE AND EXECUTE C PROGRAM FOR SERIAL TRANSMISSION AND RECEPTION USING ON-CHIP

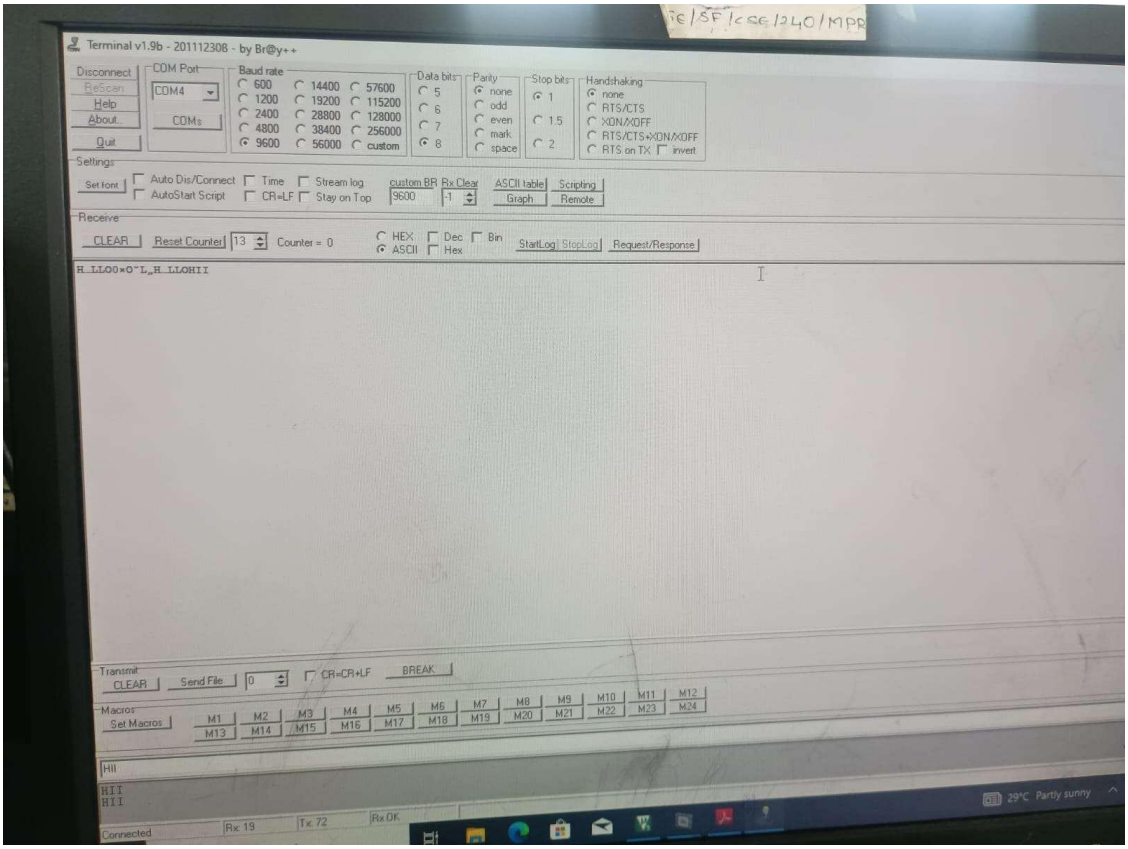
UART IN LPC2148 KIT.

APPARATUS REQUIRED:

KEIL UVISION5 SOFTWARE

PHILIPS FLAH PROGRAMMER

LPC 2148 KIT



# EXP NO: 23

## ACCESS AN INTERNAL ADC AND DISPLAY THE BINARY OUTPUT IN LEDS IN LPC2148 KIT

AIM:

TO WRITE AND EXECUTE C PROGRAM FOR ACCESSING AN INTERNAL ADC AND DISPLAY THE BINARY

OUTPUT IN LEDS IN LPC2148 KIT.

APPARATUS REQUIRED:

KEIL UVISION5 SOFTWARE

PHILIPS FLAH PROGRAMMER

LPC 2148 KIT

## EXP NO: 24

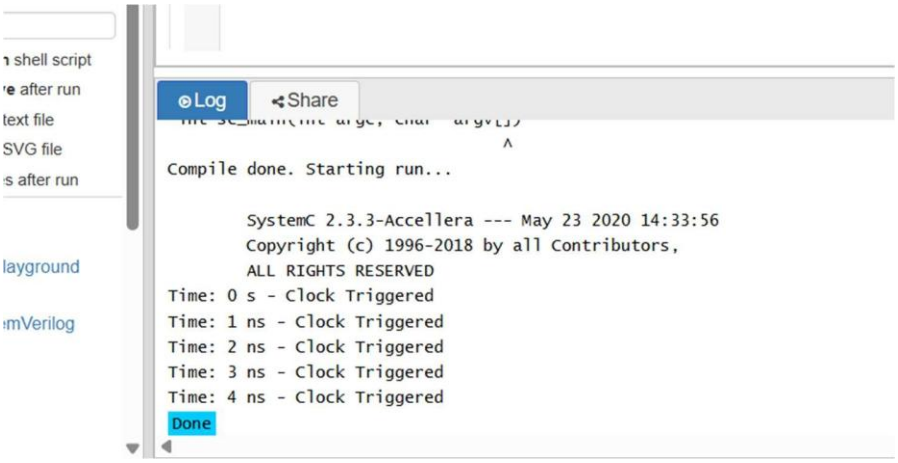
# MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

AIM:

TO UNDERSTAND AND IMPLEMENT A CLOCK-DRIVEN PROCESS USING:

- SC_CLOCK
- SC_THREAD
- WAIT()
- EVENT SENSITIVITY (POSITIVE EDGE TRIGGER)

IN SYSTEMC.



## EXP NO: 25

## PRODUCER–CONSUMER PROBLEM USING SYSTEMC

AIM:

TO IMPLEMENT AND SIMULATE THE PRODUCER–CONSUMER PROBLEM USING SYSTEMC.

SOFTWARE REQUIRED:

UBUNTU / WSL LINUX

G++ COMPILER

CMAKE

SYSTEMC LIBRARY

```cpp
#include <systemc.h>
#include "design.cpp"

int sc_main(int argc, char* argv[])
{
    // Create FIFO with capacity 5
    sc_fifo<int> fifo(5);

    // Create Producer and Consumer
    Producer producer("Producer");
    Consumer consumer("Consumer");

    // Connect FIFO
    producer.out(fifo);
    consumer.in(fifo);

    // Start simulation for 10 seconds
    sc_start(10, SC_SEC);

    return 0;
}
```

Log Share

```
        ALL RIGHTS RESERVED
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Done
```

## EXP NO: 26

## TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

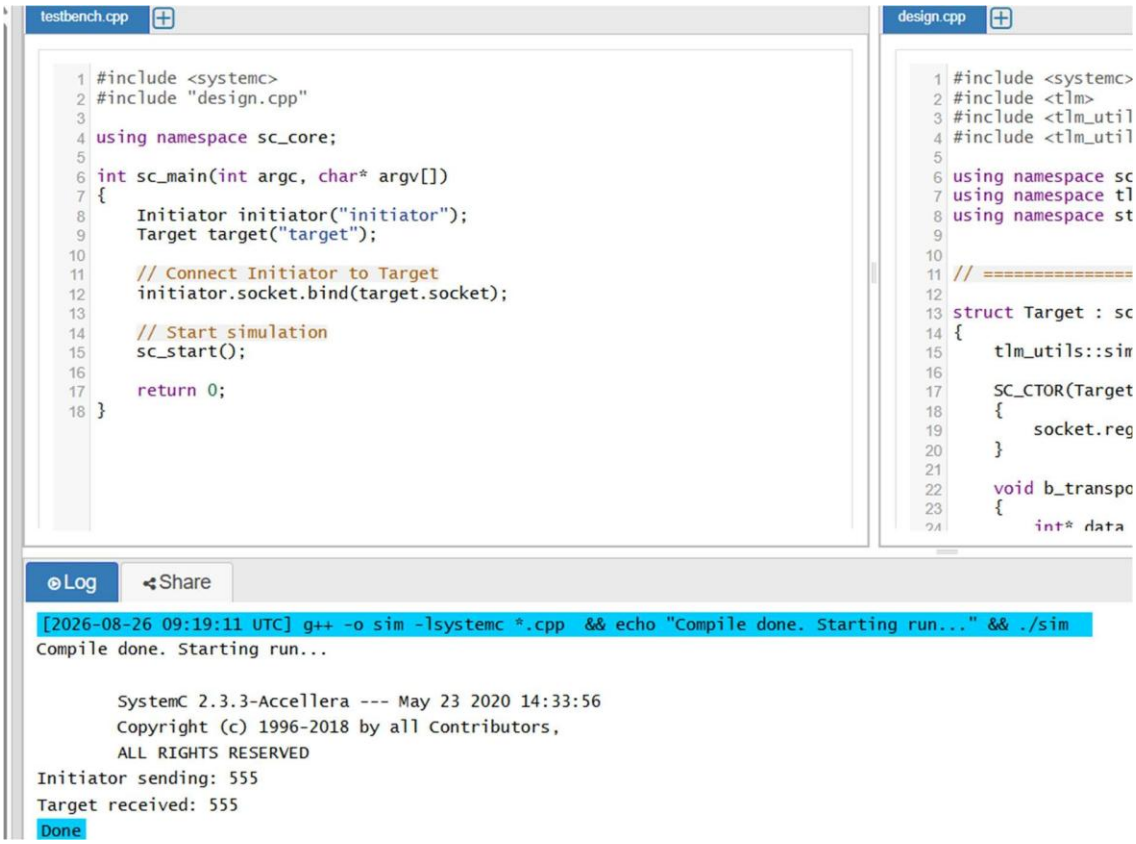
## AIM:

TO MODEL AND SIMULATE A SIMPLE TRANSACTION-LEVEL MODEL (TLM) USING INITIATOR AND

TARGET MODULES IN SYSTEMC.

SOFTWARE REQUIRED:

- UBUNTU (OR WSL UBUNTU)
- G++
- CMAKE
- SYSTEMC WITH TLM-2.0 LIBRARY



## EXP NO: 27

## BINARY SEMAPHORE IN SYSTEMC

AIM:

TO IMPLEMENT AND SIMULATE A BINARY SEMAPHORE USING SYSTEMC FOR TASK

SYNCHRONIZATION.

SOFTWARE REQUIRED:

- UBUNTU (OR WSL UBUNTU)
- G++
- CMAKE
- SYSTEMC

```
#include <systemc.h>
#include "design.cpp"

int sc_main(int argc, char* argv[])
{
    binary_semaphore obj("Binary_Semaphore");

    sc_start(10, SC_SEC);

    return 0;
}
```

```
#include

SC_MODULE
{
    int
    sc_e

    void
    {
```

Log | Share

```
[2026-08-26 09:21:31 UTC] g++ -o sim -lsystemc *.cpp  && echo "Compile done. Starting run..." && ./
Compile done. Starting run...

        SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
        Copyright (c) 1996-2018 by all Contributors,
        ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
Done
```

## EXP NO:28

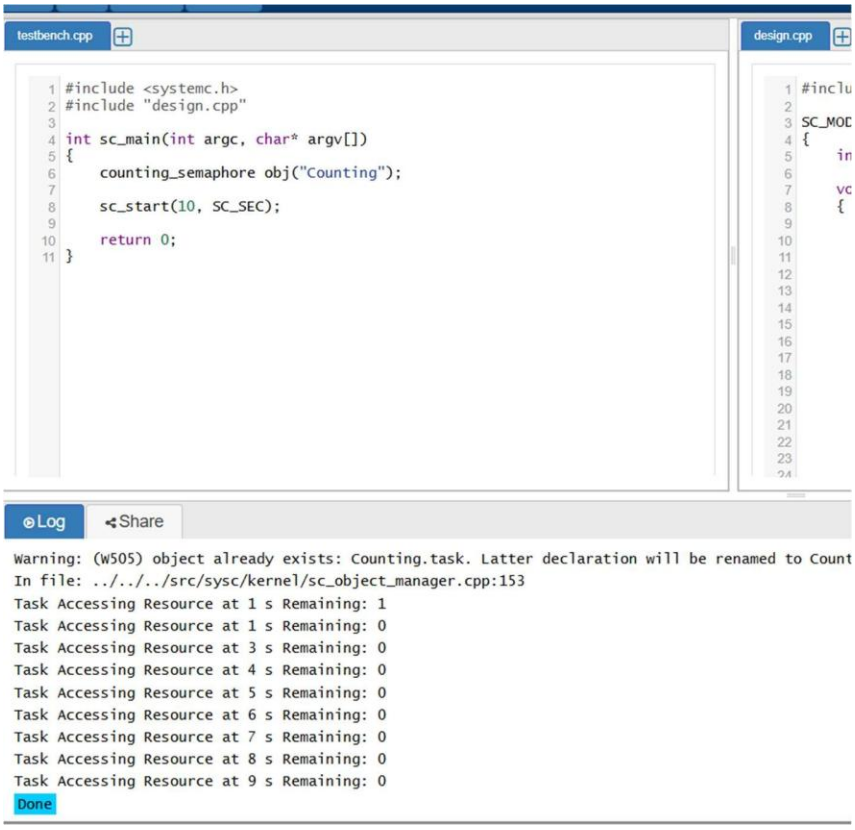
## COUNTING SEMAPHORE IN SYSTEMC

AIM:

TO IMPLEMENT COUNTING SEMAPHORE ALLOWING MULTIPLE RESOURCE ACCESS.

SOFTWARE REQUIRED:

- UBUNTU (OR WSL UBUNTU)
- G++
- CMAKE
- SYSTEMC



## EXP NO: 29

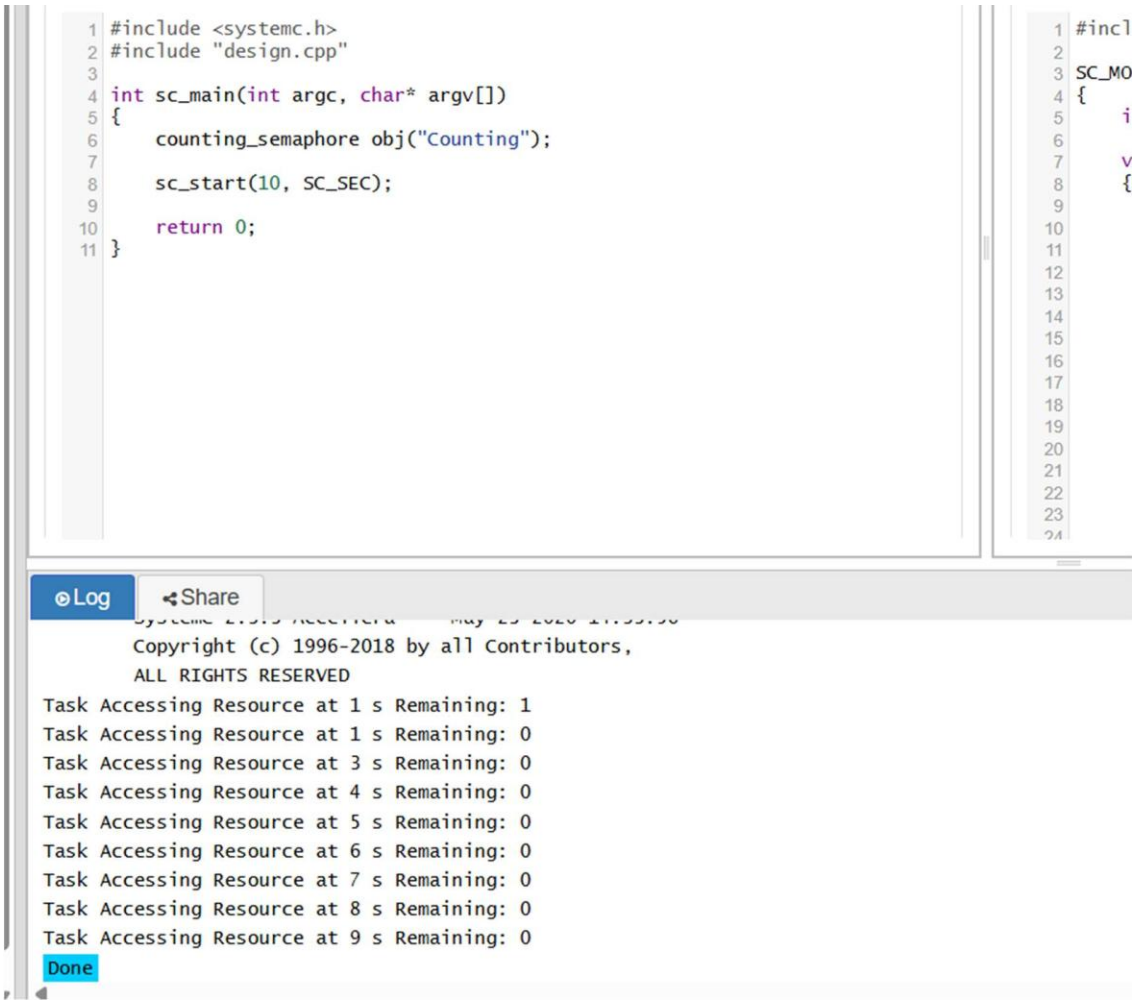
## PRIORITY-BASED INTERRUPT HANDLING

AIM:

TO MODEL PRIORITY-BASED INTERRUPT SERVICING.

SOFTWARE REQUIRED:

- UBUNTU (OR WSL UBUNTU)
- G++
- CMAKE
- SYSTEMC



```
#include <systemc.h>
#include "design.cpp"

int sc_main(int argc, char* argv[])
{
    counting_semaphore obj("Counting");

    sc_start(10, SC_SEC);

    return 0;
}
```

```
#incl

SC_MO
{
    i

    v
    {
```

Log Share

```
        Copyright (c) 1996-2018 by all Contributors,
        ALL RIGHTS RESERVED
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
Done
```

## EXP NO: 30

## RTOS TASK SCHEDULING

AIM:

TO SIMULATE ROUND-ROBIN SCHEDULING.

SOFTWARE REQUIRED:

- UBUNTU (OR WSL UBUNTU)
- G++
- CMAKE
- SYSTEMC